TODO: Title of the multimodal cancer prediction study

TODO: author names
TODO: institutions and addresses
TODO: corresponding author and email

Abstract

Background: [TODO: State the cancer prediction task, motivation, study objective, and model name.]

Results: [TODO: Summarize the completed comparisons in Tables ?? and ??, if Table ?? is retained; insert only documented results.]

Conclusions: [TODO: State conclusions supported by the completed comparisons. Do not insert a performance or novelty claim yet.]

Keywords: Multimodal learning; cancer prediction; textual data; genomic data; multimodal fusion

1 Background

[TODO: Introduce the clinical prediction task and the motivation for combining textual and genomic information.]

The Cancer Genome Atlas (TCGA) provides the data-source context for this draft [**?**]. **[TODO: Describe the specific data used and the research question.]**

The baseline reporting framework uses Table 1 of Song et al. [**?**] as the planned source of published comparison values. **[TODO: Add the study rationale and a factual description of Ours once the method has been specified.]**

2 Methods

2.1 Data and input checklist

- Cohort columns: BLCA, BRCA, GBMLGG, LUAD, UCEC, CRAD, HNSC, and STAD. **[TODO: Record cohort definitions, sample counts, inclusion criteria, and data versions.]**
- Two-modality inputs: text and genomics. **[TODO: Record data fields, preprocessing, and feature representations.]**
- Optional three-modality inputs: text, genomics, and **[TODO: third modality and its representation]**.
- Prediction target: **[TODO: survival or classification, outcome definition, labels, and prediction time point if applicable]**.

2.2 Baseline comparison checklist

- Table **??**: Cross-Attn Fusion, Early Fusion, Late Fusion, and Ours.
- Table **??**, if retained: LIMOE, MAGGate, MulT, TF, Cross-Attn Fusion, Early Fusion, Late Fusion, and Ours.

- Ours: **[TODO: model name, existing implementation identifier, and a short factual method description]**.

Bibliographic slots are reserved for Cross-Attn Fusion, Early Fusion, and Late Fusion [**?** **?** **?**], and for LIMOE, MAGGate, and Transformer Fusion (TF) [**?** **?** **?**]. MulT has a candidate original reference [**?**]. **[TODO: Resolve each TODO-CITE and confirm the mapping from each table label to its cited method.]**

2.3 Evaluation and filling checklist

- Validation for Ours: five-fold evaluation. **[TODO: Record split files, fold identifiers, and their relationship to the source study.]**
- Per-cohort reporting: fill the eight cohort columns in Table **??** and, if retained, Table **??**.
- Metric: **[TODO: full metric name, abbreviation, implementation, and whether higher or lower values are better]**.
- Summary within each cohort: **[TODO: aggregation across folds and the uncertainty measure reported beside the point estimate]**.
- Overall: **[TODO: aggregation definition, weighting, and source or calculation record for each row]**.
- Ours result records: **[TODO: paths to existing per-fold results, configurations, and the checkpoint-selection rule]**.

2.4 Provenance of comparison values

Non-Ours values in Tables **??** and **??** are planned to be transcribed from Table 1 of Song et al. [**?**]. These are literature-reported results; they are not experiments reproduced by the authors of this study. Ours rows are reserved for the authors' own experimental results.

[TODO: Record the source row, column, and setting for each transcribed value. Record the calculation used for any derived Overall entry.] All numerical entries remain unfilled in this draft.

2.5 Prototype-guided compensation for missing modalities

Scope. This subsection describes the inference-time method that is evaluated as the third strategy in Table **??**. It is a non-parametric compensation guided by the cached whole-slide image (WSI) prototypes of each patient: a missing modality is filled with features taken from the most similar training patient, shrunk towards the training mean by a setting-specific coefficient, and compensation can be switched off for a setting. Nothing is trained in this study. The survival networks are frozen (one previously trained NPJ-C model per cohort and seed, 25 in total), and the method involves no learnable prototypes, no learned compensation module, no moving-average update, no contrastive or alignment loss and no additional training objective. The experiment covers five TCGA cohorts (BLCA, BRCA, LGG, LUAD and UCEC), three modalities and five seeds; the seeds index independently trained networks and are not cross-validation folds. The equations below restate the retrieval and filling steps of the executed computation and are not a derivation of optimality.

Modality availability. Let c denote a cohort, i a query patient, j a candidate donor, $m \in \{W, R, T\}$ the WSI, RNA and text modalities, g an evaluation setting and $\mathcal{I}_c$ the training patients of cohort c. The cached feature array of modality m before projection is $\mathbf{X}_i^m$. A modality is visible to the model ($v_{i,g}^m = 1$) if and only if it is naturally available and not masked by setting g; the settings `rna_100`, `text_100` and `both_100` mask RNA, text and both of them, respectively, `none` adds no artificial mask, and the WSI is never masked. Naturally missing modalities remain missing, so `none` is not a complete-data condition. The modalities to be compensated are $\mathcal{M}_{i,g} = \{m \in \{R, T\} : v_{i,g}^m = 0\}$.

Retrieval key from valid WSI prototypes. The WSI cache $\mathbf{P}_i = \mathbf{X}_i^W \in \mathbb{R}^{128\times 1536}$ holds at most 128 rows per patient, namely patient-level cluster centres of patch embeddings or, when fewer rows are available, the available rows followed by zero padding; it is used as stored and no clustering is repeated. The retrieval key averages the valid (non-zero) rows and centres the result,

$$\mathbf{q}_i = \frac{\sum_{k=1}^{128} a_{ik}\,\mathbf{P}_{ik}}{\sum_{k=1}^{128} a_{ik}} - \boldsymbol{\mu}_c^W, \qquad a_{ik} = \mathbf{1}\big[\mathbf{P}_{ik} \neq \mathbf{0}\big], \tag{1}$$

where $\boldsymbol{\mu}_c^W$ averages the first term over the training patients of cohort c with equal weight per patient. WSI zero padding therefore never enters the retrieval key, and validation and test patients never contribute to $\boldsymbol{\mu}_c^W$.

Similarity, candidates and donor selection. The similarity between query i and training patient j is

$$s_{ij} = \cos\big(\mathbf{q}_i, \mathbf{q}_j\big) + J_{i,g}\,\alpha\,\cos\big(\bar{\mathbf{r}}_i, \bar{\mathbf{r}}_j\big), \tag{2}$$

where cos denotes cosine similarity, $\bar{\mathbf{r}}_i$ is the token mean of $\mathbf{X}_i^R$ without centring and $\alpha = 1$. The indicator $J_{i,g}$ equals one only if $g =$ `text_100`, features are retrieved in that setting ($\gamma_{c,g} = 1$ and $\lambda_{c,g} = 1$, both defined below FIXME!!!) and the patient's own RNA is naturally available; otherwise $J_{i,g} = 0$ and no RNA is read. The donor is the most similar admissible training patient,

$$j^\star = \arg\max_{j \in \mathcal{D}_{i,g}} s_{ij}, \tag{3}$$

where $\mathcal{D}_{i,g}$ is the set of training patients of cohort c, other than i, that naturally possess every modality in $\mathcal{M}_{i,g}$, and also both RNA and text when $J_{i,g} = 1$. When both RNA and text are missing and features are retrieved, one donor therefore supplies both. Validation and test patients never enter the patient bank, and survival times and censoring indicators are not used for retrieval.

Compensated value. With $\boldsymbol{\mu}_c^m$, $m \in \{R, T\}$, the element-wise mean of modality m over the training patients of cohort c that possess it, the compensated value is

$$\widetilde{\mathbf{X}}_i^m = \boldsymbol{\mu}_c^m + \lambda_{c,g}\big(\mathbf{X}_{j^\star}^m - \boldsymbol{\mu}_c^m\big), \qquad \lambda_{c,g} \in \{0, 1\}. \tag{4}$$

The WSI prototypes act only as the retrieval key: the value returned is the donor's complete feature array of the target modality, and cluster centres are never substituted for RNA or text. The coefficient $\lambda_{c,g}$ is specific to cohort c and setting g. With $\lambda_{c,g} = 1$ the compensated value equals the donor's original features; with $\lambda_{c,g} = 0$ it reduces to the training mean and no retrieval is performed.

Filling and fusion mask. A binary gate $\gamma_{c,g} \in \{0, 1\}$, also specific to cohort c and setting g, determines whether compensation is applied:

$$\mathbf{X}_{i,\text{in}}^{m} = \begin{cases} \widetilde{\mathbf{X}}_{i}^{m}, & v_{i,g}^{m} = 0 \text{ and } \gamma_{c,g} = 1, \\ \mathbf{X}_{i}^{m}, & \text{otherwise.} \end{cases} \tag{5}$$

With $\gamma_{c,g} = 0$ the missing modality is marked invalid and excluded from fusion; with $\gamma_{c,g} = 1$ the filled modality is marked valid. Both reference strategies are special cases of this formulation, namely no compensation ($\gamma_{c,g} = 0$ throughout) and mean imputation ($\gamma_{c,g} = 1$ and $\lambda_{c,g} = 0$ throughout). The constant α and the default $\lambda_{c,g} = 1$ were selected on the validation split. **[TODO: Author to add: values of $\lambda_{c,g}$ and $\gamma_{c,g}$ for each cohort and setting (needed to reproduce the identical entries of Table ??).]**

3 Results

3.1 Compensation strategies under missing modalities

Table **??** compares three inference-time strategies for missing modalities on identical frozen networks: no compensation, mean imputation, and the proposed prototype-guided compensation defined in Section **??**. Averaged with equal weights over the five cohorts, five seeds and four settings, the proposed method reached a macro-average concordance index (C-index) of 0.640506, compared with 0.631824 for mean imputation and 0.630751 without compensation. In the present evaluation the proposed method thus increased the average discriminative performance. Values below are given in the order proposed method, mean imputation, no compensation. The cohort-level overall value was higher than both references in four of the five cohorts (BRCA, LGG, LUAD and UCEC), although the margin in LUAD was small (0.587039 versus 0.587004 and 0.586206), and it was not higher in BLCA (0.578982 versus 0.582191 and 0.583388). This count refers to the overall averages and does not mean that the proposed method was best in every setting. The larger differences occurred when both auxiliary modalities were masked, in BRCA (0.640263 versus 0.614738 and 0.596026), UCEC (0.609348 versus 0.575072 and 0.576319) and LGG (0.665286 versus 0.635266 and 0.576190), and for UCEC in the text-masked setting (0.609748 versus 0.577743 and 0.589272). When only RNA was masked, or without additional masking, the three strategies were close, and mean imputation or no compensation was highest in several cohorts. No statistical test was performed. **[TODO: Author to add: limitation concerning how the LUAD setting was selected.]**

3.2 Two-modality comparison

[TODO: Fill Table ?? from the designated sources and the existing Ours results, then describe the observed per-cohort and Overall comparisons.]

3.3 Three-modality comparison: optional

[TODO: Confirm whether Table ?? is retained and specify the third modality. If retained, fill its entries and describe the comparisons.]

4 Discussion

[TODO: Interpret the completed comparisons without inventing a mechanistic explanation or claiming an unmeasured improvement.]

[TODO: Discuss the findings in relation to the cited methods, distinguishing literature-reported values from the authors' results.]

Table 1: Comparison of three compensation strategies across five cancers under four modality-availability settings. Entries are test C-indices averaged over five seeds; *Overall* is the equally weighted mean of the four settings of a cohort. Proto. denotes the proposed prototype-guided compensation (Section **??**).

	BRCA			UCEC			LUAD			BLCA			LGG		
Scenario	No comp.	Mean	Proto.	No comp.	Mean	Proto.	No comp.	Mean	Proto.	No comp.	Mean	Proto.	No comp.	Mean	Proto.
RNA masked	0.683842	0.684751	**0.685705**	0.647229	**0.655464**	0.651413	**0.580660**	0.580239	0.580239	**0.593546**	0.589105	0.588168	0.767219	**0.790959**	0.786611
Text masked	**0.655002**	0.619121	**0.655002**	0.589272	0.577743	**0.609748**	**0.596630**	0.595156	**0.596630**	**0.575989**	0.574913	**0.575989**	**0.705866**	0.664113	**0.705866**
Both masked	0.596026	0.614738	**0.640263**	0.576319	0.575072	**0.609348**	0.585469	**0.588803**	**0.588803**	0.574046	**0.574774**	0.561797	0.576190	0.635266	**0.665286**
None	**0.691359**	0.686068	0.688112	0.651502	**0.656888**	0.650968	0.582064	**0.583819**	0.582485	**0.589972**	**0.589972**	**0.589972**	0.796825	**0.799517**	0.797723
Overall	0.656557	0.651170	**0.667270**	0.616081	0.616292	**0.630369**	0.586206	0.587004	**0.587039**	**0.583388**	0.582191	0.578982	0.711525	0.722464	**0.738872**

Scenarios: RNA masked, Text masked and Both masked remove the RNA profile, the pathology report or both for all test patients (`rna_100`, `text_100`, `both_100`); None (`none`) adds no artificial mask but retains naturally occurring missingness and is therefore not a complete-data condition. Overall is an average and not a fifth experimental scenario. Strategies: No comp., no compensation (m0real), missing modalities are masked; Mean, mean imputation (m1), a missing modality is replaced by the element-wise training mean of the same cohort; Proto., the proposed prototype-guided compensation (strategy0; Section **??**). Bold marks the highest unrounded value among the three strategies within a cohort and row; ties are all bold. The three strategies use the same 25 frozen networks; no model was retrained and no statistical test was performed.

Table 2: Two-modality comparison using text and genomics. Metric and uncertainty summary: TODO.

Method	BLCA	BRCA	GBMLGG	LUAD	UCEC	CRAD	HNSC	STAD	Overall
Cross-Attn Fusion	–	–	–	–	–	–	–	–	–
Early Fusion	–	–	–	–	–	–	–	–	–
Late Fusion	–	–	–	–	–	–	–	–	–
Ours	–	–	–	–	–	–	–	–	–

Non-Ours values are planned to be taken from Table 1 of Song et al. [**?**], not reproduced in this study. Ours values will be supplied from the authors' own experiments. The symbol – denotes an unfilled entry. Overall: TODO, including whether it is transcribed or calculated.

Table 3: Optional three-modality comparison using text, genomics, and a third modality (TODO). Metric and uncertainty summary: TODO.

Method	BLCA	BRCA	GBMLGG	LUAD	UCEC	CRAD	HNSC	STAD	Overall
LIMOE	–	–	–	–	–	–	–	–	–
MAGGate	–	–	–	–	–	–	–	–	–
MulT	–	–	–	–	–	–	–	–	–
TF	–	–	–	–	–	–	–	–	–
Cross-Attn Fusion	–	–	–	–	–	–	–	–	–
Early Fusion	–	–	–	–	–	–	–	–	–
Late Fusion	–	–	–	–	–	–	–	–	–
Ours	–	–	–	–	–	–	–	–	–

Non-Ours values are planned to be taken from Table 1 of Song et al. [**?**], not reproduced in this study. Ours values will be supplied from the authors' own experiments. TF denotes Transformer Fusion. The symbol – denotes an unfilled entry. Overall: TODO, including whether it is transcribed or calculated.

[TODO: State the limitations supported by the final study design and completed evidence.]

5 Conclusions

[TODO: Summarize the findings and their scope after the result tables have been completed.]

Abbreviations

TCGA The Cancer Genome Atlas.

TF Transformer Fusion.

Cross-Attn Cross-Attention.

[TODO: Add the full names of the reported metric, cohort labels, and remaining method abbreviations; confirm the definition of CRAD.]

Declarations

Ethics approval and consent to participate

[TODO: Author-confirmed ethics and consent statement, including approval, exemption, or applicability details.]

Consent for publication

[TODO: Author-confirmed consent-for-publication statement.]

Availability of data and materials

[TODO: Dataset access details and links to available code, split files, and result records. State any actual access restrictions.]

Competing interests

[TODO: Author-confirmed declaration of competing interests.]

Funding

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Authors' contributions

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